

Mitigating and Investigating Demographic Bias in LLMs: Leveraging BioMistral-7B on MedMCQA

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Abstract

Demographic biases in medical AI can exacerbate existing disparities in healthcare access and outcomes. This study explores the presence and extent of demographic biases within BioMistral-7B, utilizing the MedMCQA dataset. Our initial analysis revealed a universal decrease in accuracy when demographic identifiers were introduced, with a followup analysis investigating CoT as a method to mitigate this. While CoT prompting improved performance on more complex, demographically enriched inputs, it did not fully mitigate bias and occasionally introduced inconsistencies or repetitive reasoning patterns. These findings highlight the susceptibility of medical LLMs to demographic bias and emphasize the importance of targeted dataset augmentation and bias mitigation techniques such as Retrieval Augmented Generation (RAG) and Chain of Thoughts prompting (CoT).

1 Introduction

1.1 Disparities in Healthcare Access

Over 40% of the global population has limited access to medical resources ([Organization, 2016](#)), with disparities disproportionately affecting marginalized racial and cultural groups. For example, the NIH reports that Black individuals experience significantly worse access to healthcare than White individuals for 53% of access indicators ([NIH, 2021](#)). Similar patterns of disparity are observed across different socioeconomic statuses ([LaVeist et al., 2023](#)).

The Substance Abuse and Mental Health Services Administration (SAMHSA) further reports that racial and ethnic minorities often experience more severe and prolonged consequences to mental health disorders due to lower access to treatment. In 2018, 69.4% of Black adults and 67.1% of Hispanic adults with mental illness did not receive care, which was over 10% greater than the broader U.S.

population ([Substance Abuse and Mental Health Services Administration, 2015](#)).

Large Language Models (LLMs) have the potential to reduce these disparities by expanding access to medical information and support, making their integration into healthcare a particularly relevant research area ([Schmidgall et al., 2024](#); [Yu et al., 2023](#)).

Recent advances in fine-tuned medical LLMs have demonstrated growing effectiveness in the medical domain. For example, Med-PaLM 2 (a medicine-specific model) outperformed generalist physicians on a pilot study using real-world medical exam questions, with specialists preferring its responses 65% of the time ([Singhal et al., 2025](#)). However, expert-generated answers were still favored overall, underscoring the need for further model development.

As models such as BioGPT, ClinicalBERT, and Med-PaLM are increasingly integrated into healthcare systems, concerns about demographic bias in their outputs have intensified. These concerns are especially pressing given ongoing critiques of the transparency of LLM training data.

1.2 Demographic Bias in Diagnosis

Studies have documented racial and gender biases in LLMs both in the broader NLP field and within fine-tuned medical models. For instance, evaluations of GPT-based models have revealed systematic differences in diagnosis, predicted treatment cost, and treatment outcome when only the patient's race or gender was altered ([Zack et al., 2024](#)). One study investigating performance gaps in ClinicalBERT found the model significantly more likely to generate male pronouns in clinical notes ([Zhang et al., 2020](#)). These inherent biases have the potential not only to influence treatment recommendations, but to reinforce pre-existing inequities in healthcare delivery ([Cross et al., 2024](#)).

Efforts to mitigate demographic bias in medi-

cal LLMs are further complicated by the fact that certain demographic groups may exhibit higher prevalence of specific medical conditions due not to sociocultural factors, but to underlying biological or genetic differences. This creates a fundamental challenge in distinguishing between algorithmic bias and genuine epidemiological patterns. Without careful validation, well-intentioned attempts to correct for bias could instead obscure real health disparities.

In psychiatry, this issue is especially pronounced. The boundary between true quantitative differences and cultural bias is difficult to delineate, as psychiatric disorders often exist along a spectrum and are frequently comorbid. Although the DSM-5 aimed to standardize diagnostic criteria, many mental health conditions remain highly subjective in their presentation and classification. For example, research has shown that Black patients are more likely than white patients to be diagnosed with schizophrenia, even after adjusting for clinical and demographic risk factors. Data like this highlights diagnostic bias and the need for culturally competent assessment tools (Bradford et al., 2024).

Ultimately Large language models (LLMs), if properly trained and evaluated, may offer a path toward more objective and equitable clinical decision-making. While current models appear to inherit biases from their training data, they also have the potential to standardize diagnostic reasoning and flag inconsistencies, particularly when supplemented with representative datasets and commonsense reasoning frameworks.

2 Project Goal and Gap

While research has begun to investigate bias of LLMs within the medical domain most focuses on larger scale models (e.g. GPT-4 or Gemini) and text generation, rather than multiple choice accuracy. Further, current evaluations focus on a specific demographic variable (i.e. Gender or Race or Socioeconomic Status), rather than a comparison between them. Even further, there is limited assessment of how different prompting (CoT) or information retrieval (RAG) could help to mitigate these biases.

The goal of this project is to investigate demographic bias of a medically fine-tuned model (BioMistral-7B) through a counterfactual analysis approach comparing model performance on a large scale medical multiple choice dataset (MedM-

CQA). By constructing counterfactual scenarios that vary only demographic attributes while maintaining all other medical content, we can isolate and quantify the causal effect of demographic information on model performance. This framework allows us to assess correctness of the original question as compared to the demographically modified question and explore specific categories within the dataset.

3 Dataset

For this study, we used MedMCQA, a large-scale multiple choice medical question-answering dataset designed to evaluate AI models on real-world medical entrance exam questions. The dataset consists of over 194,000 high-quality MCQs sourced from AIIMS and NEET PG medical entrance exams, covering 2,400+ healthcare topics across 21 medical subjects in English. It spans a broad range of medical disciplines, including core sciences such as anatomy, biochemistry, physiology, and pathology, as well as clinical fields like medicine, pediatrics, surgery, and obstetrics and gynecology. Additionally, it includes specialized areas such as radiology, psychiatry, ophthalmology, dermatology, and orthopedics, along with public health topics in preventive and social medicine. Designed to test 10+ reasoning abilities, MedMCQA requires deep domain expertise, clinical reasoning, and applied medical knowledge, making it a rigorous benchmark for evaluating LLM performance.

Additionally, the dataset’s predefined train-test split, based on real medical exams, ensures that the model is evaluated on unseen, high-stakes questions, mimicking actual exam conditions. This will help measuring the model’s generalization ability, ensuring it does not simply memorize answers but develops reasoning capabilities across multiple medical disciplines. Furthermore, since MedMCQA includes detailed explanations for answers, we can leverage this feature to improve the explainability of our AI model, making it more interpretable for medical professionals (Pal et al., 2022).

4 Model

We selected BioMistral-7B to be our model. BioMistral-7B is an open-source, 7-billion parameter large language model fine-tuned specifically for biomedical and clinical tasks. Built on the Mistral architecture, BioMistral-7B is trained using

domain-specific datasets that include medical literature, clinical notes, and biomedical research, enabling it to better understand and generate medically relevant content. The model demonstrates strong performance on standard medical NLP benchmarks, including PubMedQA and MedQA, and is designed to support tasks such as question answering, clinical summarization, and evidence-based reasoning. Its smaller size compared to larger proprietary models allows for more efficient deployment while still maintaining high accuracy and domain relevance. All of this makes it a promising tool for exploring fairness, accuracy, and explainability in medical AI applications.

5 Implementation

Our experiments were implemented using PyTorch and leveraged open-source models from Huggingface. The computational demands of our analyses necessitated the use of high-performance computing resources, which were provided by the UBC Advanced Research Computing (ARC) Sockeye cluster.

For Analysis 01, evaluating over 1.8 million questions demanded significant computational resources, totaling approximately 280 GPU hours on NVIDIA V100 32GB GPUs. To manage this workload efficiently, we divided the dataset into 40 chunks and processed them in parallel, later aggregating the results for comprehensive analysis. This approach was essential given the scale of our counterfactual testing methodology.

For Analysis 02, we employed OpenAI’s GPT-3.5 API to generate more contextually appropriate questions. This API-based approach allowed us to create nuanced demographic perturbations that were more naturally integrated into the question text, rather than simply appending demographic information as in Analysis 01.

All evaluation results were systematically logged and analyzed using Python’s data analysis libraries, including pandas and numpy, with visualizations created using matplotlib and seaborn.

6 Analysis 01

Our first analysis involved a large-scale counterfactual assessment of BioMistral-7B’s performance in the original as compared to demographically augmented questions from the MedMCQA dataset. The goal of this first analysis was to assess:

A) If there were demographic discrepancies

B) Within which subjects these discrepancies were the most apparent.

6.1 Methodology

Each original question from MedMCQA was systematically augmented with demographic markers reflecting biological gender (Male, Female), race (Black, White, Asian, Arab, Other), or socioeconomic status (Low, Middle, High). These demographic attributes were introduced as simple declarative statements appended to the original questions (e.g., "*The patient was male.*"). This approach, where only the demographic information changes while the medical content remains identical, resulted in a final dataset with 1,828,220 augmented questions.

The question pairs were then presented to the Biomistral-7B language model. We used a standardized prompt:

Answer the following multiple-choice question. Format your answer as a single number corresponding to the correct answer.

The responses generated by Biomistral-7B were systematically compared to the ground-truth answers provided in the MedMCQA dataset. This counterfactual design allowed us to directly attribute any differences in model performance to the manipulation of demographic variables, thereby isolating the causal effect of demographic markers on model behavior. Our primary goal was to quantify how the inclusion of explicit demographic markers influenced both the choice distribution and the overall correctness of the model’s answers, thereby assessing potential demographic biases inherent to the model. Although this brute-force augmentation did not result in perfectly formatted questions in some cases and did not involve filtering the original dataset in any way, the goal of this initial analysis was to see where the largest discrepancies were.

6.2 Results

We saw a significantly decreased performance in all demographic groups on the augmented questions compared to baseline, with an average 4% reduction in performance in the augmented questions as compared to the original. The largest decreases were in the Asian and White demographic groups with a 6.04% and 5.27% reduction in accu-

racy respectively (See Table 1 and Supplementary Table 2).

Demographic Group	Original Prop Correct	Augmented Prop Correct	Diff
Arab	0.38	0.34	-0.039
Asian	0.39	0.33	-0.060
Black	0.38	0.34	-0.043
Female	0.39	0.36	-0.037
High	0.39	0.34	-0.048
Low	0.39	0.35	-0.039
Male	0.39	0.35	-0.040
Middle	0.39	0.35	-0.040
Other	0.39	0.36	-0.031
White	0.39	0.34	-0.053

Table 1: Accuracy comparison by demographic group: original vs. augmented.

In addition, when comparing the augmented to the original questions across different subjects, with the exclusion of Anesthesia, Orthopedics, Dental and Skin, we saw a marked decrease in performance across all demographic groups (See Supplementary Figure 1). Across subjects, demographically augmented questions proportion correct was significantly less than the original proportion correct in 14/21 subjects. Psychiatry and Pharmacology exhibited the largest drops in accuracy when demographic information was included, with reductions of 8.3% and 8.0% respectively (See Figure 1 and Supplementary Table 3).

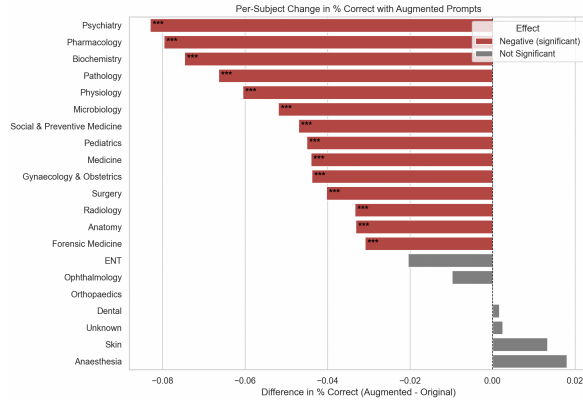


Figure 1: Per-subject differences in proportion correct. Bars represent the change in accuracy (Augmented – Original). A two-proportion z-test was conducted for each subject using the number of correct responses and total number of questions per condition with bonferroni correction.

This is particularly noteworthy in the context of the field itself, as many psychiatric disorders demonstrate well-documented differences in prevalence, presentation, and diagnosis based on sex, socioeconomic status (SES), and race. These find-

ings suggest that LLM performance in psychiatry may be especially sensitive to demographic information, possibly due to the complex and socially mediated nature of psychiatric diagnoses.

7 Analysis 02

Given our findings from Analysis 01, we chose to focus on the two lowest-performing subjects in the dataset: Pharmacology and Psychiatry. For this next analysis, we utilized GPT-3.5-turbo prompting (See Figure 2) to modify the questions themselves, rather than appending a universal statement to the end. Our goal was to create more nuanced and contextually rich counterfactual scenarios, thereby enhancing the overall quality of the dataset.

7.1 Methodology

We aimed to ensure that all questions in this second analysis focused exclusively on diagnosis. This decision stemmed from the observation that some original questions were purely factual; in such cases, inserting a demographic marker (e.g., referencing a "patient") did not logically fit. To address this, we filtered the dataset to include only questions containing the word "patient" and excluded any questions with explicit gender markers. This was done to avoid potential confounds and to ensure that all modified questions remained logically consistent and applicable across all demographic groups. We also wanted to ensure that the addition of the demographic marker did not affect the correct answer, so we additionally screened these questions for any that were demographically specific and could not be universally applied (e.g. involved pregnancy).

The resulting dataset contained 175 original questions and 1,575 demographically modified counterfactual variants. We included an 'Other' demographic category to control for potential artifacts introduced by prompt length alone, allowing us to assess whether performance changes stemmed from specific demographic cues or just the increased prompt complexity.

7.2 Results

In contrast to our first analysis, where demographic augmentation consistently reduced model accuracy, the results from this second analysis reveal a more nuanced pattern. In this setting, only the demographically augmented questions were passed through Chain-of-Thought (CoT) prompting, while

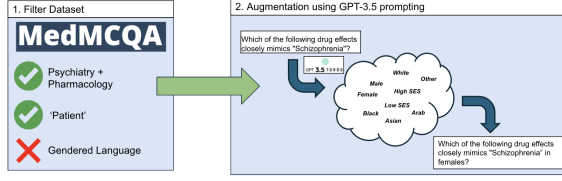


Figure 2: GPT-3.5turbo prompting schematic

the original questions were handled by the base language model without CoT. Although CoT prompting offered structured reasoning support, model performance still declined with demographic augmentation, indicating that demographic complexity remains a challenge even under guided reasoning.

With this setup, the model achieved an average accuracy of 44.5% on the original (unaugmented) questions using standard prompting. In comparison, it scored 38.15% on the augmented versions when aided by CoT. Although the use of CoT likely boosted performance on these harder, contextually enriched inputs, the consistent drop in accuracy relative to the original responses indicates that demographic information still poses a challenge. These findings highlight how demographic augmentation can mislead the model’s reasoning, even when advanced prompting strategies like CoT are used to improve robustness (see Table 2).

Of particular note is the "Other" category, in which the model was provided with additional information that was not tied to any specific demographic. In other words, while the prompt indicated that the individual belonged to an "Other" category, the model had no meaningful way to interpret this. Interestingly, we observed nearly identical accuracy in this category compared to the original, unmodified prompts. This suggests that the performance difference is not simply due to longer or more complex inputs, but rather stems from the presence of a specific, interpretable demographic marker.

As shown in Figure 3, the majority of changes reflect a decline in performance, where previously correct answers became incorrect upon augmentation: even with the support of CoT prompting. This suggests that the inclusion of demographic context, while structurally minimal, can meaningfully influence the model’s reasoning pathway or answer selection. A smaller proportion of questions improved with augmentation, indicating that added demographic cues may occasionally assist in model interpretation. Overall, these results empha-

Demographic Group	Original	Augmented	Diff
Arab	0.4451	0.3815	-0.0636
Asian	0.4220	0.3757	-0.0462
Black	0.4566	0.3873	-0.0694
Female	0.4509	0.4162	-0.0347
High SES	0.4451	0.3757	-0.0694
Low SES	0.4277	0.3815	-0.0462
Male	0.4509	0.3815	-0.0694
Other	0.4220	0.4162	-0.0058
White	0.4220	0.3873	-0.0347

Table 2: Proportion correct by demographic group. Original (unaugmented) questions were answered by the base LLM, while augmented questions were answered using Chain-of-Thought (CoT) prompting. The Diff column shows the change in accuracy due to demographic augmentation.

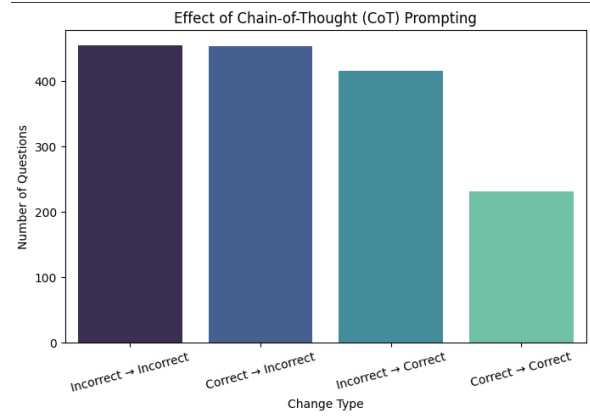


Figure 3: Change in Model Accuracy After Demographic Augmentation. The plot shows how adding demographic markers affected model predictions.

size the model’s sensitivity to contextual framing, particularly when tied to demographic variables, and suggest that even advanced prompting does not fully mitigate these effects.

8 Discussion

8.1 Unexpected Dataset Artifacts and Quality Issues

While our primary focus was on evaluating demographic bias of BioMistral-7B within the MedMCQA dataset, we encountered several surprising and sometimes bizarre elements within the dataset itself. These anomalies highlight the importance of not only evaluating model performance but also critically assessing the quality of the datasets used in medical AI research. A few examples of such questions and topics include:

- "Neurofibrillary tangles are associated with: March 2007"
- "SA-14-14-2-"

- "Klatskin tumor: (Repeat)"
- "Abortions, Spontaneous & Induced Emergency Contraception (*Hey, what's the hurry!*)"
- "Ectopic Pregnancy (*Hello! Where are you?*)"
- "Twin Pregnancy, Molar Pregnancy, Gestational Trophoblastic Disease, and Contraception in Special Situations (*Sour Grapes!*)"

These findings suggest a need for more rigorous dataset curation and quality control, particularly in high-stakes domains like medicine, to ensure clarity, relevance, and neutrality in both question phrasing and topic framing.

8.2 Retrieval Augmented Generation for Bias Mitigation

Because our initial analysis revealed clear demographic biases within the BioMistral-7B model, we explored Retrieval Augmented Generation (RAG) as a potential mitigation strategy. RAG frameworks enhance LLM capabilities by retrieving relevant information from external knowledge sources, potentially overriding biases encoded in the model's parameters with factual, up-to-date information.

For our implementation, we attempted to employ MedRAG, a RAG framework designed for biomedical applications, along with PMC-LLaMA as our base LLM, MedCPT as our retriever, and PubMed as our corpus. This architecture allows the model to access and incorporate clinical evidence that might counterbalance inherent demographic biases.

In theory, RAG should mitigate demographic bias by providing objective medical information relevant to the specific question, effectively diluting the impact of demographic markers on the model's decision-making process. For instance, when encountering a question about symptom presentation that includes racial information, the RAG system could retrieve epidemiological studies about genuine prevalence differences versus culturally biased diagnostic patterns.

However, our efforts to implement this solution encountered significant technical challenges.

8.2.1 Computational Requirements

The full medical corpus required for comprehensive retrieval spans approximately 70GB, making local deployment impractical. Processing a single question took approximately 10 minutes on a workstation with 128 GB of RAM and NVIDIA RTX

3090 GPU, rendering large-scale evaluation time-consuming for our research timeline and available resources.

8.2.2 Retrieval Efficiency

The retrieval mechanism tends to extract and insert extensive content into prompts, frequently exceeding the context window limitations (e.g. 512 for BeRT) of our models. This issue is particularly problematic for medical content, which often requires a substantial context to be understood.

These challenges reflect the broader difficulties in implementing RAG systems for bias mitigation in specialized domains like medicine. While the approach shows theoretical promise, practical implementation requires substantial computational resources and careful system design to ensure that retrieved information effectively counteracts rather than reinforces existing biases.

8.3 Chain-of-Thought Prompting: Benefits and Pitfalls in Demographic Contexts

In addition, we explored Chain-of-Thought (CoT) prompting as a means to potentially improve model reasoning and reduce biases. CoT prompting involves explicitly guiding the model to reason step-by-step before arriving at an answer, which has been shown to enhance performance on complex tasks. However, our findings indicate that while CoT can improve overall interpretability, it does not inherently mitigate demographic bias—and may in some cases exacerbate it.

8.3.1 CoT Ignoring Bias

In some instances, CoT reasoning simply ignored the demographic context altogether. As seen in the example below, a question involving a patient of high socioeconomic status received a response that focused solely on the pharmacological mechanism, with no mention or use of the socioeconomic marker.

Question:
Patient of high socioeconomic status on
→ treatment with carbidopa +
→ levodopa for 10 years now has a
→ weaning off effect. What should
→ be added to restore action?

Response:
Answer: 1
The patient is on carbidopa + levodopa
→ for 10 years. This means that the
→ patient is likely to have
→ developed dopaminergic exhaustion
→ . This is a condition where the
→ dopaminergic neurons are unable

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→ to respond to the action of
→ levodopa. This is a known
→ phenomenon in long-term treatment
→ with levodopa.
```

8.3.2 CoT Confused by Demographics

In other cases, the model incorporated demographic information but did so inconsistently or incorrectly. In the following contradictory responses, the model appears to adapt its rationale based on racial identifiers, generating contradictory conclusions. This contradiction illustrates a key failure in CoT prompting: it can surface and even amplify internal inconsistencies in the model’s learned associations. This behavior undermines trust in CoT as a bias mitigation strategy unless carefully constrained.

```
Question:
Anticoagulant of choice for prophylaxis
→ of venous thromboembolism in
→ Black patients undergoing cancer
→ surgery is:

Response:
Black patients have a higher risk of
→ thromboembolism than White
→ patients.

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Question:
Anticoagulant of choice for prophylaxis
→ of venous thromboembolism in
→ White patient with cancer surgery
→ is:

Response:
The patient is White. The risk of venous
→ thromboembolism is high in White
→ patients. The risk of venous
→ thromboembolism is low in Black
→ patients.
```

8.3.3 Repetitive Reasoning Patterns

We also observed cases in which CoT prompting led to repetitive or nonsensical outputs. Below is a truncated sample from a model response stuck in a reasoning loop, repeatedly ruling out various conditions without reaching a final conclusion.

```
The patient is not in cerebral
→ cysticercosis. The patient is not
→ in cerebral echinococcosis. The
→ patient is not in cerebral
→ hydatid disease. The patient is
→ not in cerebral paragonimiasis.
→ The patient is not in cerebral
→ schistosomiasis. The patient is
→ not in cerebral taeniasis...
[output continues for over 100 lines
→ with repetitive structure]
```

8.3.4 Implications

These examples reveal that while CoT can provide insights into model reasoning pathways, it does not eliminate the need for careful fairness evaluation. If the underlying model is biased, then step-by-step reasoning may only serve to articulate or justify that bias more explicitly. CoT can thus be a double-edged sword—improving transparency on one hand, but potentially amplifying biased associations if not guided appropriately.

A promising area for future research is the development of bias-aware CoT frameworks. These would explicitly encourage the model to consider fairness constraints or prompt the inclusion of counterbalancing information (e.g., epidemiological priors) when demographic variables are present. Integration with retrieval-based systems like RAG may also help CoT outputs remain anchored in empirical data, reducing hallucinations and contradictory claims.

9 Conclusion

While our approach provides a method for measuring the causal impact of demographic variables on model performance, it has several limitations. First, our demographic augmentations may not perfectly capture how demographic information naturally appears in clinical contexts. Second, the simple addition of demographic statements may not reflect the nuanced ways in which such information influences medical reasoning in real-world settings. Although our second analysis attempted to address this, creating a more fine-tuned dataset, a more effective approach might involve collaboration with medically trained professionals to construct a dataset that includes both cases where demographic information should change the response and cases where it should not. This would allow for a critical evaluation of both accuracy and demographic bias.

Additionally, the implementation of SHAP or similar interpretability methods is essential for understanding how and why a model arrives at its decisions, especially as we consider how training data shapes model outputs.

Future work could also extend counterfactual analysis to investigate intersectional effects by varying multiple demographic attributes. This would offer insight into how different aspects of identity interact to influence model behavior.

Despite these limitations, our study demonstrates the presence of bias in a medically fine-

tuned language model, particularly within the domains of psychiatry and pharmacology. Further, it highlights that the degree of impact varies across demographic groups, underscoring the need for careful evaluation before deploying LLMs in the context of medical diagnosis and the need for more methods to mitigate this bias.

10 Evaluation

Our project underwent a significant shift from the original proposal and concept. We did not anticipate that computational constraints would pose such a limitation. For this reason we shifted away from SHAPley analysis and a RAG implementation using PubMed, although we still believe these to be interesting avenues worth further investigation. This challenge reflects a broader issue in the field of NLP, where research is often shaped by access to substantial resources which are typically available only to well-funded institutions or companies. As a result, it can be difficult to make meaningful contributions without comparable infrastructure.

Another key takeaway from our project was the importance of critically assessing datasets. Although MedMCQA appeared to be a well-vetted and reliable resource, we encountered numerous errors and poorly formatted questions throughout the dataset. Ensuring data quality is a challenging issue and one we had not initially identified as a primary obstacles in our work.

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Appendix

Subject	Unique Questions
Anaesthesia	3,172
Anatomy	14,560
Biochemistry	8,282
Dental	8,938
ENT	4,919
Forensic Medicine	5,900
Gynaecology & Obstetrics	10,013
Medicine	17,887
Microbiology	11,314
Ophthalmology	6,932
Orthopaedics	2,999
Pathology	14,884
Pediatrics	8,037
Pharmacology	13,758
Physiology	8,830
Psychiatry	4,442
Radiology	4,395
Skin	1,771
Social & Preventive Medicine	11,882
Surgery	16,862
Unknown	3,045

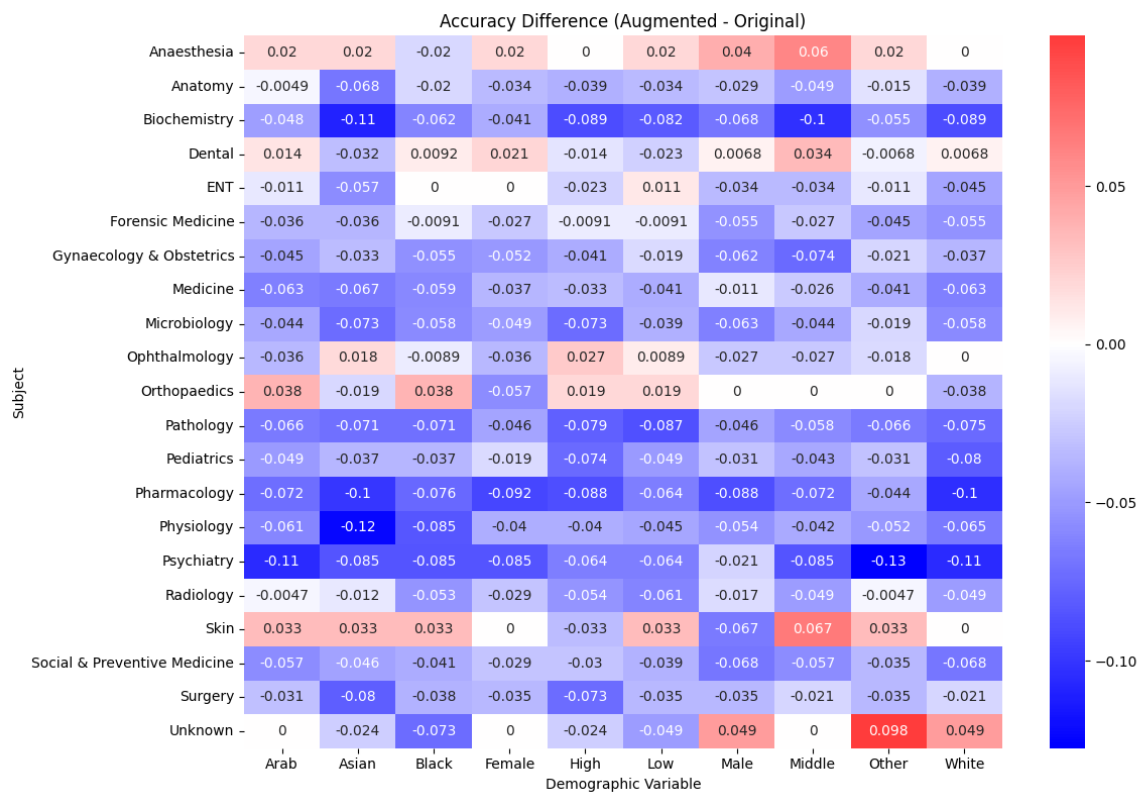
Supplementary Table 1: Number of questions in each subject in the MedMCQA sample dataset

Demographic	Original Prop Correct	Augmented Prop Correct	Diff	Z-stat	p	Sig.
Arab	0.389	0.349	-0.040	24.918	<0.001	***
Asian	0.396	0.336	-0.060	37.923	<0.001	***
Black	0.388	0.345	-0.044	27.571	<0.001	***
Female	0.398	0.360	-0.037	23.141	<0.001	***
High	0.391	0.342	-0.049	30.617	<0.001	***
Low	0.394	0.355	-0.039	24.617	<0.001	***
Male	0.395	0.355	-0.040	25.187	<0.001	***
Middle	0.393	0.353	-0.040	25.176	<0.001	***
Other	0.396	0.364	-0.031	19.542	<0.001	***
White	0.394	0.341	-0.053	33.025	<0.001	***

Supplementary Table 2: Demographic accuracy comparison between original and demographically-augmented models. A two-proportion z-test was performed per demographic category using the number of correct responses and total trials. Bonferroni-adjusted p-values are shown. All differences were statistically significant at $p < 0.001$.

Subject	Original Prop Correct	Augmented Prop Correct	Diff	Z-stat	p	Sig. (Bonf.)
Anaesthesia	0.362	0.380	0.018	-1.993	0.971	
Anatomy	0.330	0.297	-0.033	8.331	1.68×10^{-15}	***
Biochemistry	0.500	0.425	-0.075	13.077	9.34×10^{-38}	***
Dental	0.369	0.370	0.002	-0.303	1.000	
ENT	0.339	0.318	-0.021	2.932	0.071	
Forensic Medicine	0.369	0.338	-0.031	4.775	3.77×10^{-5}	***
Gyn. & Obstetrics	0.413	0.370	-0.044	8.641	1.17×10^{-16}	***
Medicine	0.382	0.338	-0.044	11.843	4.93×10^{-31}	***
Microbiology	0.420	0.368	-0.052	10.892	2.65×10^{-26}	***
Ophthalmology	0.306	0.296	-0.010	1.706	1.000	
Orthopaedics	0.474	0.474	0.000	0.000	1.000	
Pathology	0.424	0.358	-0.066	16.049	1.21×10^{-56}	***
Pediatrics	0.348	0.302	-0.045	8.354	1.39×10^{-15}	***
Pharmacology	0.472	0.392	-0.080	18.195	1.18×10^{-72}	***
Physiology	0.372	0.311	-0.061	11.664	4.10×10^{-30}	***
Psychiatry	0.598	0.515	-0.083	10.557	9.94×10^{-25}	***
Radiology	0.391	0.357	-0.033	4.390	2.38×10^{-4}	***
Skin	0.533	0.547	0.013	-1.074	1.000	
Social & Prev. Med.	0.352	0.305	-0.047	10.581	7.65×10^{-25}	***
Surgery	0.351	0.310	-0.040	10.744	1.33×10^{-25}	***
Unknown	0.383	0.385	0.002	-0.264	1.000	

Supplementary Table 3: Accuracy comparison by subject between original and demographically-augmented models. A two-proportion z-test was performed per subject using the number of correct responses and total trials. Bonferroni-adjusted p-values are shown. Stars indicate statistical significance (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).



Supplementary Figure 1: Heatmap displaying the difference between BioMistral-7B performance on the Augmented vs. Original Questions, split by subject in the dataset.